

Research Protocol

Study Title : Architecture Guidelines for In-company Platform and Technology Departments: A Sustainability-Driven Preliminary Study

Aim

This study aims to establish an evidence-based foundation for a Statement of Architectural Work (SoAW) for architecture guideline for compute that enables enterprise technology departments particularly in regulated financial institutions to operationalize sustainability as a first-class non-functional requirement within compute architectures.

Research Questions:

RQ1: How can compute architecture guidelines be defined and implemented to improve sustainability in enterprise technology departments?

RQ2: Which architectural principles must be operationalized to embed sustainability into the compute lifecycle of high-availability, regulated environments?

Methodology:

A systematic mapping study complemented by qualitative expert interviews.

Data Sources:

Academic Literature (2016-2025): ACM Digital Library, IEEE Xplore, SpringerLink

Empirical Industry Data: Internal architectural standards, ESG frameworks, anonymized expert interviews.

Expert Interviews

Participants: Solution Architects, Platform Engineers, Enterprise Architects, Product Managers

Sampling: Purposive, role-based

Format: Semi-structured (45-60 minutes)

Ethics: Fully anonymized, no PII

Search Strategy (Replicability)

IEEE Xplore: ("architecture" OR "architectural guideline") AND ("compute" OR "cloud-native" OR "platform engineering") AND ("sustainability" OR "energy efficiency" OR "carbon-aware")

ACM Digital Library: ("architecture guideline" OR "software architecture") AND ("sustainable compute" OR "energy-efficient cloud" OR "carbon-aware computing")

SpringerLink: "architecture" AND "compute sustainability" AND ("cloud" OR "enterprise" OR "platform")

Selection Criteria

Inclusion: 2016-2025, architectural focus, energy or carbon metrics, enterprise context

Exclusion: Pre-2016, hardware-only, non-English, ESG policy without technical depth

Outputs:

Sustainable architecture classification framework and methodological basis for a formal SoAW